

Mingwei Ma

Guangzhou, Guangdong, China

+86 13935155531 | mmm593860159@gmail.com

Nationality: Chinese | **Current Position:** Master's Student



Profile

Master's student in Control Science and Engineering at South China University of Technology. Research interests include interactive 6D object modeling, pose estimation and tracking, robotic perception, and robotic manipulation, with experience in computer vision, 3D reconstruction, and learning-based robotics.

Education

South China University of Technology

Academic Master's Candidate, Control Science and Engineering

Sep 2025 – Present

Guangzhou, China

- School of Automation Science and Engineering

Zhengzhou University

Bachelor of Engineering in Automation

Sep 2021 – Jun 2025

Zhengzhou, China

- School of Electrical and Information Engineering

Research Experience

Interactive Click-to-Model and SPARK-6D

2025 – 2026

Research Project, South China University of Technology

- Led the development of an interactive click-to-model pipeline that estimates an object's 6D pose from a user click in an image. Extended the framework with SPARK-6D through more robust re-tracking and accelerated tracking inference.

Automated Sealant Inspection System

2025 – 2026

Research Project, South China University of Technology

- Developed an automated sealant inspection system integrating multi-view imaging, 3D reconstruction, and defect localization for aircraft sealant surfaces.

Dual-Arm Robotic System for Transmission-Line Maintenance

2025 – 2026

Research Project, South China University of Technology

- Developed a dual-arm robotic system for removing foreign objects from overhead power transmission lines.

Publications

Accepted

- *Click-to-Model: Real-Time Interactive Object Modeling and Robust 6D Pose Tracking.*

Mingwei Ma, Yu Ren, Lunshuo Tian and Yang Cong

2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026), accepted.

An interactive framework for constructing object models from user input and maintaining robust real-time 6D pose estimates.

Manuscripts Under Review

- *SEAMap: Reconstruction-Guided Robotic Active Inspection via Unified Surface Mapping for Sealant Surfaces.*

Mingwei Ma, Yu Ren, Supeng Diao, and Yang Cong.

2027 AAAI Conference on Artificial Intelligence (AAAI 2027), under review.

An automated pipeline for automating the inspection of sealant using a robotic arm equipped with a sleeve.

- *GeoSeal: Joint Multimodal Representation and 3D Reconstruction for Reflective Sealant Inspection.*

Yu Ren, Mingwei Ma, Supeng Diao, Baihao Huang, and Yang Cong.

IEEE Transactions on Multimedia (TMM), JIF: 9.7, Q1; under review.

A coarse-to-fine inspection framework combining regional video screening, multimodal 3D reconstruction, spatial defect alignment, and mirror-assisted inspection of reflective sealant surfaces.

- *Safety-Oriented Command Regulation for Assisted Teleoperation in Overhead Transmission-Line Maintenance.*

Hua Wang, Yu Ren, Mingwei Ma and Yang Cong.

2027 IEEE International Conference on Robotics and Automation (ICRA 2027), under review.

A training-free framework combining SE(3) state prediction, risk-aware neural refinement, optical-flow gating, and GPU

execution optimization for real-time 6D object tracking.

Honors and Competitions

11th Robotic Grasping and Manipulation Competition (RGMC)

Runner-up, Cloud Robotics Track

Jun 2026

IEEE ICRA 2026

Team members: Shenghan Huang, **Mingwei Ma**, Jiahao Long, Jiawei Weng, and Yang Cong

• Represented South China University of Technology in a CloudGripper-based robotics challenge focused on remote robotic grasping and manipulation.

• Collaborated with team members to develop, test, and demonstrate a competition-ready robotic manipulation solution.

Skills

Research Areas	Interactive object modeling; 6D object pose estimation and tracking; robotic perception; manipulation; control science and automation
Research Methods	Literature review; experimental evaluation; quantitative analysis; academic writing; technical presentation; team-based research
Languages	Mandarin Chinese (native); English (professional working proficiency)